STAGE 1

PROBLEM DISCOVERY REPORT [PDR]

Team ID: V26-GF-101

Hardware Based/Software Based/Other: SOFTWARE

Title of the Project: WariSetu

Track in which your Project fits in: Open Innovation Track

Name of the Team Leader: Prince Vyas

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1. The Problem

The Pandharpur Wari draws 27-28 lakh pilgrims into a town of 1.4 lakh residents within a single 20-hour window (Ashadhi Ekadashi 2025). Three problems repeat every year. **Family separation:** 2,700+ missing-person reports in 2025, ~98% resolved in 24-48 hours, but only through manual "Missing Cell" outposts, loudspeakers and printed photos; 50-60 cases still need days of active tracing. **Delayed medical response:** 108/MEMS attended 553 emergencies in 2017 and treated ~50,956 pilgrims on-site; heart attacks, heat exhaustion and highway gridlock keep delaying care, once leaving a burn patient waiting 30+ minutes for an ambulance. **Disconnected infrastructure:** per the Solapur District Disaster Management Plan, police, health, irrigation and municipal departments run separate systems, and the one integrated command centre (500+ cameras, 6 AI drones) covers only Pandharpur town, not the 250 km transit corridor where recorded fatalities (Jejuri, Panvel, Pandharpur outskirts) have actually occurred. The gap isn't infrastructure. It's intelligence to connect what already exists.

2. Solution, USP & Key Features

WariSetu is a software layer on top of the CCTV network, control rooms and toll-free helplines the administration already runs, no app, wristband or registration needed. Three functions run continuously: Crowd Intelligence (live density/flow flags at choke points before a crush forms), Voice Lost & Found (a Marathi call is converted into an automatic CCTV face search, returning the last-seen location within minutes), and Medical Risk Detection (the camera feed flags falls, collapses and heat distress for the nearest medical team). All three feed one command-centre dashboard.

This is the only zero-adoption layer of its kind, reaching elderly and low-literacy pilgrims an app never would, and the only system that turns a spoken description into an automated CCTV face search, which doubles as an anti-pickpocketing layer (75 gangs caught in a single recent Wari). The same platform also handles route diversion, resource dispatch, highway-corridor monitoring and post-event analytics, and reuses directly at Nashik Kumbh Mela 2027, Sabarimala, Rath Yatra, stadiums and metro stations, since only the camera feed changes between deployments.

3. Technology, Feasibility & Cost

Stack: YOLOv11 + ByteTrack and InsightFace/ArcFace for vision, Whisper/IndicWhisper for Marathi speech, FastAPI + Node.js backend, React/Next.js + MapLibre dashboard, PostgreSQL/Redis/Qdrant for data, Docker/Kubernetes on AWS/GCP, Exotel/Twilio for IVR, all open-source or off-the-shelf. Solapur's 2025 operation already runs 500+ cameras, six AI drones and an integrated control room, so this is orchestration, not new hardware; the Solapur DDMP itself calls for exactly this kind of centralized monitoring framework.

Component	Pilot (3-week)	Full-scale (annual)
CV / GPU inference	₹40k - 60k	₹25 - 40 lakh
IVR voice lost & found	₹20k - 30k	₹10 - 15 lakh
DB & vector search	₹8k - 12k	₹6 - 10 lakh
Dashboard + AMC	In-house	₹40 - 60 lakh + AMC
Total (indicative)	₹70k - 1 lakh	₹1.5 - 2.5 crore

For reference, Solapur separately sought ~₹1.5-2 crore for a comparable AI crowd system, so this band sits within demonstrated government appetite to fund it.

4. Impact, Business Model & Risks

Impact: near-real-time face search instead of days of manual tracing; faster medical alerts; monitoring extended to the highway corridor where fatalities actually occur; one reporting layer for district audits. **Revenue:** a government SaaS licence (~₹15-25 lakh/district/year) plus AMC, CSR co-funding from foundations already active at the Wari (Finolex/Mukul Madhav, Vithalrao Vikhe Patil), and multi-event licensing to other pilgrimages. **Risks:** face search on crowded, low-light footage will throw false positives, so a control-room officer always confirms a match before dispatch; rural stretches of the corridor have patchy connectivity, so heavy processing runs on-device and only relevant clips are sent; and on privacy, face embeddings are stored only for an active case and purged once it's resolved, with no biometric database retained between seasons.

5. Conclusion

Pandharpur already has the cameras, ambulances and control rooms; what it lacks, by its own disaster management plan's admission, is a layer that connects them. WariSetu closes that gap without new infrastructure, at a cost already inside what the district has budgeted for comparable systems, and it's built to travel: the same pipeline moves to Nashik Kumbh Mela 2027 without redesign.

Images / Screenshots

(space reserved for UI mockups, architecture diagram and dashboard screenshots)